



Proposed CEA Link — December 2026

A 24-line link built to the owner's final rules: the ticket office rostered as fixed pairs, every other duty chosen so the floor follows the December 2026 timetable

Prepared 28 September 2026 · built for the December 2026 timetable and assessed by the workspace's own rule modules (Marylebone Roster v24.35) · figures on this page are computed, not typed

PROPOSAL

Final Rules

The owner's final rules — the ticket office fixed, the floor fitted to the timetable

CODE **FR-24-F34**

FINGERPRINT **03a59c77**

BUILT FROM duty table F · seed 34 · 24 lines · 4 cover weeks

- ✓ **24** lines designed
- ✓ **35h 00m** a week — exactly the contract
- ✓ **0** hard-limit breaches
- ✓ **0** rests under 12h
- ✓ **0** fatigue factors present (today: 5)

Five questions a manager asks first, in order. Every figure is computed from the cells; **page 2 says how to read each one**, with today's link and the rest of the folder marked on a scale.

CAN IT BE RUN?

Yes

the hard limits

0 rests under 12h · longest run 6 of 13 · contract exact

DOES IT MEET THE DECEMBER SHAPE?

4 of 4

headcount rules met

at least four to open, three to close and five at 22:00, every day; 14 and 10 at the weekend · today meets 0 of 4

HOW TIRING IS IT?

0

fatigue factors present, of 25

advisory — present means worth a look, not a breach · today 5 · fewest in the folder 0

DOES IT FOLLOW THE TRAINS?

21.9

weekday demand fit — lower is closer

Saturday 12.2 · Sunday 33.6 · today 51.1 · best in the folder 30.7 · the floor alone 20.4 · 11.6 · 19.2

IS IT FAMILIAR?

5 of 17

shift times people work today

12 new, listed on page 5 · 11 of 20 weeks are one turn (today 7 of 16)

6 in 24

full weekends off

25% of the rotation, against 20% today (4 in 20) · most in the folder 25%

4 cover weeks

lines 1, 7, 13, 19

evenly spaced six lines apart — today's sit at 1, 7, 12, 17

1h 50m a week

typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

What this is

A proposal for the CEA link once the December 2026 timetable arrives, for a team of 24, built to the owner's final rules of 28 September 2026. The ticket office is written in first, as fixed pairs of identical turns — Monday to Friday two 06:20–14:20 and two 14:00–22:30, Saturday two 06:20–14:50 and two 14:30–22:00, Sunday two 07:15–15:30 and two 13:30–22:30 for an office open 07:15–22:30 — and **those four people are not counted as floor cover**. Every weekday closer starts 15:45.

Everything else was chosen for one thing: how closely the people on the floor follow the trains. The headcounts are minimums, not targets — at least four at the open, three at the close and five at 22:00 on every day, 14 on a Saturday and 10 on a Sunday — with at least two on the floor at every moment, a 15-minute handover to each closer and 20 minutes in the ticket office, Sunday duties of 8h to 9h, and no more shift times than today's 18. Each day's table was enumerated to a proof rather than sampled. Page 6 shows the cover three ways: everyone on duty, the ticket office, and the floor.

What it is not. Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep. The 24-line length, the four cover weeks and the December 2026 headcounts are owner-relayed figures with no document behind them (evidence class C), and the 13-day limit's policy citation is still outstanding — pages 7 and 8 say so where it applies. It clears every hard limit the tool can check — twelve hours between duties, the 13-day limit and the exact contract — which is the floor for being worth discussing.

HOW TO READ THE NUMBERS

What each figure means, and what good looks like

Every figure on this sheet is computed from the cells. On each scale: this sheet in navy, today's 20-line link as a ring, the best and worst of the other 22 sheets in the folder as gold and grey ticks.

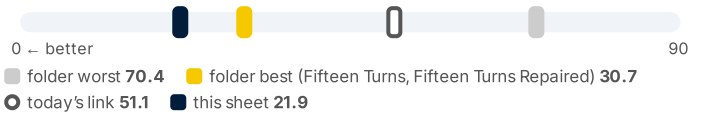
Three things to hold. Lower is better on the fit and the fatigue count; higher is better on rest, weekends, one-turn weeks and familiarity. "Present" is not "failed" — the ORR list reports what is there. Today's link is the yardstick, not the target: it meets none of the December 2026 headcounts.

DOES IT FOLLOW THE TRAINS?

Demand fit — weekday

How closely the people on duty follow the trains through an average weekday. For each of its 18 hours, the hour's share of the staff-minutes is set against its share of the traffic; the gaps, in percentage points, are squared and added up — 0 would be a perfect match. The typical hourly gap is the square root of the score ÷ 18: about 1.1 points here, 1.7 today.

What it does not tell you. It compares shares, not headcount — adding people evenly to every hour changes nothing — and the station rosters posts that need someone whether the hour carries 23 trains or 5, so a flatter day can be right and still score worse.



Saturday **12.2** (today 45.7, folder best 8.1) · Sunday **33.6** (today 65.9, best 20.9). Each day type has its own curve and window, so a Saturday is only compared with a Saturday.

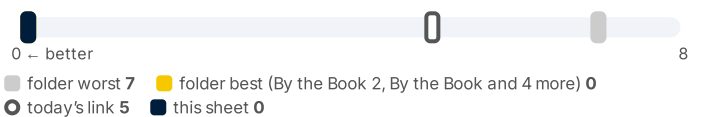
On page 6.

HOW TIRING IS IT?

Fatigue factors present

How many of the 25 roster patterns on the Office of Rail and Road's good-practice list appear anywhere in the rotation — long runs of earlys, short gaps between duties, start times that jump — counted by the workspace's own module.

What it does not tell you. A factor present is worth a look, never a breach: the ORR says the list is guidance, not limits. Zero does not mean approved and one does not mean unsafe; page 8 names which, and how big.



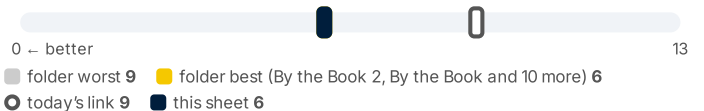
On page 8.

CAN IT BE RUN?

Longest run of worked days

The most consecutive days anyone works in the worst case — a cover week's four duties placed as badly as they can be. Chiltern's limit is 13; the workspace's own target is 6.

What it does not tell you. The roster clerk decides where cover-week duties fall, so the as-rostered figure can be lower; page 8 shows both readings.



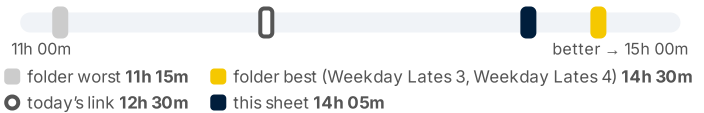
On page 7.

CAN IT BE RUN?

Tightest rest between duties

The shortest gap anywhere between one duty ending and the next beginning — Saturday into Sunday and the last line into the first included. Twelve hours is the floor; a design under it cannot be run.

What it does not tell you. It is one gap. A rotation with many 12h05 rests and one with none read the same here; the count under 12h is in the chips on page 1.



On page 7.

WHAT IS IT LIKE TO WORK?

Full weekends off

The share of weeks with both Saturday and Sunday off. Today's link gives 4 in 20, one weekend in 5; this sheet gives 6 in 24, one weekend in 4.

What it does not tell you. A weekend off beside a 42-hour week is still a weekend off; the hours by week are on the grid, page 4.



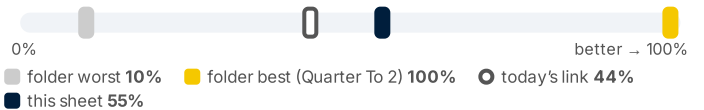
On page 5.

WHAT IS IT LIKE TO WORK?

Working weeks on one turn

The share of working weeks where a person keeps the same clock time all week. A week that mixes an early and a late is what people mean by "all over the place".

What it does not tell you. One turn all week can still be a bad week if the turn is long; the week-to-week move and the duty lengths are on pages 5 and 8.



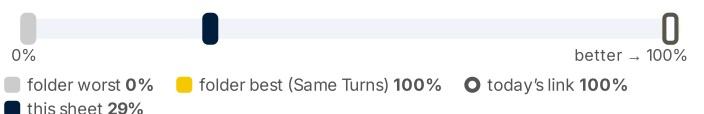
On page 5.

IS IT FAMILIAR?

Shift times people work today

The share of this sheet's clock times that somebody on the 20-line link already works. It is the folder's proxy for how much there is to learn and to accept.

What it does not tell you. Familiar is not the same as popular — a time everyone works and nobody likes still counts — and a new time in one week is not the same as one in ten. The duty table on page 5 lists the new ones.



On page 5.

BEFORE THE FIGURES

The decision this sheet supports

What is being decided, which criteria are hard and which are the room's, how much each figure can bear, and what no page here can tell you. Read this before the tiles.

- 1

What this is
- 2

How to read the numbers
- 3

The decision it supports (this page)
- 4

The 24-line grid, with the words explained
- 5

A week compared with today
- 6

People on duty, hour by hour
- 7

Hard limits and the December 2026 figures
- 8

The ORR fatigue factors
- 9

Where it came from
- 10

The grid, ready to paste into the app

The question

Should the link for the December 2026 timetable be built to the owner's final rules — the ticket office rostered as fixed pairs and kept off the floor, 15:45 closers, the headcounts as minimums — with every other duty chosen so the floor follows the trains, accepting the 12 times nobody works today that this costs?

Final Rules is the owner's final brief, answered. It is read against **Pinned Turns 2 (P2-24-N13)** and **Pinned Turns (PT-24-P34)**, the two earlier briefs, and **By the Book (BB-24-D7)**, the rules-first answer — not against every design in the folder.

The criteria, in three tiers

HARD — A DESIGN MEETS THESE OR CANNOT BE RUN

- ✓ Twelve hours between duties — met, tightest 14h 05m
- ✓ Chiltern's 13-day limit — longest run 6 days
- ✓ The 35-hour contract, exactly — 35h 00m

SOFT — THE ROOM WEIGHS THESE AGAINST EACH OTHER

- The owner's headcount minimums met: **4 of 4** (today 0 of 4)
- Fatigue factors present: **0** (today 5)
- Demand fit weekday · Sat · Sun: **21.9** · 12.2 · 33.6 (today 51.1 · 45.7 · 65.9; lower is closer) — the floor alone 20.4 · 11.6 · 19.2, the figure the tables were chosen on
- Times worked today: **5 of 17** · full weekends off: **6** in 24 (today 4 in 20)

PREFERENCE — STAFF'S TO STATE, NOT THE TOOL'S

- Early against late: earlies 7h 25m–9h 30m, lates 7h 00m–9h 00m
- 12 of 17 clock times are new — the duty table on page 5 lists them
- The longest duty, 9h 30m, and who holds it
- Cover weeks at lines 1, 7, 13, 19

How much to trust each figure

FIGURE ON THESE PAGES	WHERE IT COMES FROM	WEIGHT
0 rests under 12h · 35h 00m contract · 14/10 at the weekend	Computed from the cells; re-computed by the workspace on import	Firm
Longest run 6 days, inside 13	Computed; the 13-day limit's policy citation is still outstanding	Firm figure , unconfirmed threshold
At least four at the open, three to close, five at 22:00, 14 and 10 at the weekend	Owner-relayed December 2026 headcounts with no document behind them (class C)	Provisional
0 fatigue factors present	The ORR's 2021 list, which it states are not limits; 4 factors still "definition to confirm"	Advisory only — never pass or fail
Demand fit 21.9 · 12.2 · 33.6	Measured timetable, one share-based measure on one December 2026 timetable curve	Indicative
5 of 17 times are today's	Computed against the live 20-line link	Firm , but a proxy for acceptability
Search score 12700	The annealer's own objective; page 9 explains it	Do not weigh

What no page here can tell you

- Whether the people who would work them accept a table in which 12 of 17 times are new (page 5 lists them) — familiarity is measured as "times worked today", which is a proxy for acceptability, not acceptability.
- Whether the roster clerk splits a cover week day-on-day-off. It is the one thing that moves the 48-hour-break figure (9 as a block, 9 split), and the link does not decide it.
- How leave and sickness cover land on a 24-line link with four cover weeks (lines 1, 7, 13, 19); nothing here models absence.
- Whether the December 2026 timetable demand curve holds on Wembley event days — Saturday was fitted to the timetable evenly, with no weighting for events, and the curve is a measured timetable, not a footfall count.

Before it is frozen

Lines 2, 3, 4, 6, 8 hold the times nobody works today; the people who would work those weeks read them first and say in one line each what they would change. The roster office answers the cover-week question. Someone confirms the source of the 13-day limit. Only then does the factor count on page 8 mean anything, and only then is this sheet more than an aid to a conversation.

THE ROTATION

The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

LINE	SUN	MON	TUE	WED	THU	FRI	SAT	MON-SAT hours	ALL WEEK hours	DAYS WORKED Mon-Sat	TURNS Mon-Fri
1	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
2	13:30 22:30	14:00 22:30	RD	RD	07:00 16:15	07:00 16:15	06:20 14:50	35h 30m	44h 30m	4	2
3	07:15 15:30	06:20 14:20	RD	RD	06:20 14:20	07:00 16:15	06:20 15:30	34h 25m	42h 40m	4	2
4	RD	RD	06:20 14:30	07:00 16:15	06:20 14:30	06:20 14:30	06:20 15:30	42h 55m	42h 55m	5	2
5	RD	RD	06:20 14:20	06:20 14:20	06:20 14:20	06:20 14:20	RD	32h 00m	32h 00m	4	1
6	RD	06:20 13:45	06:20 13:45	06:20 13:45	RD	RD	07:15 16:45	31h 45m	31h 45m	4	1
7	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
8	RD	RD	15:45 23:55	15:45 23:55	15:45 23:55	15:45 23:55	RD	32h 40m	32h 40m	4	1
9	RD	07:00 16:15	06:20 15:45	06:20 15:45	06:20 15:45	06:20 15:45	RD	46h 55m	46h 55m	5	2
10	RD	06:20 14:30	07:00 16:15	06:20 14:30	07:00 16:15	RD	06:20 15:30	44h 00m	44h 00m	5	2
11	07:15 15:30	06:20 14:20	06:20 14:20	RD	RD	06:20 14:20	07:15 16:45	33h 30m	41h 45m	4	1
12	07:15 15:30	07:00 16:15	07:00 16:15	07:00 16:15	RD	RD	14:30 22:00	35h 15m	43h 30m	4	1
13	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
14	09:15 18:15	RD	RD	06:20 14:20	06:20 13:45	06:20 13:45	06:20 14:50	31h 20m	40h 20m	4	2
15	RD	RD	14:00 22:30	14:00 22:30	14:00 22:30	14:00 22:30	14:45 22:45	42h 00m	42h 00m	5	1
16	13:30 22:30	RD	RD	14:00 22:30	14:00 22:30	14:00 22:30	RD	25h 30m	34h 30m	3	1
17	RD	14:00 21:00	14:00 21:00	14:00 21:00	RD	RD	15:45 23:55	29h 10m	29h 10m	4	1
18	15:15 23:25	15:45 23:55	15:45 23:55	RD	RD	15:45 23:55	14:30 22:00	32h 00m	40h 10m	4	1
19	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
20	07:15 15:30	06:20 15:45	RD	RD	14:00 21:00	14:00 21:00	15:45 23:55	31h 35m	39h 50m	4	2
21	15:15 23:25	15:45 23:55	RD	RD	15:45 23:55	15:45 23:55	RD	24h 30m	32h 40m	3	1
22	RD	15:45 23:55	15:45 23:55	15:45 23:55	15:45 23:55	15:45 23:55	RD	40h 50m	40h 50m	5	1
23	RD	14:00 22:30	14:00 22:30	15:45 23:55	15:45 23:55	RD	15:45 23:55	41h 30m	41h 30m	5	2
24	15:15 23:25	15:45 23:55	15:45 23:55	15:45 23:55	RD	RD	15:45 23:55	32h 40m	40h 50m	4	1
Cover	10 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	35h 00m	38h 31m	4.20	11 one turn
Minutes	5,070	6,970	6,970	6,970	6,970	6,970	7,150	42,000	47,070		

Early turns Late turns Cover (spare) week — four duties of seven, any turn Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines · Turns: distinct clock times Monday to Friday, **bold** where the week is one turn (one clock time Mon-Fri and one early-or-late family across every worked day — the page 1 figure) · Minutes: duty minutes per day, and Mon-Sat is the contract (20 × 35h = 42,000)

The words on this page

Link — the whole rota: 24 weeks laid out as 24 lines. Everyone works line 1, then line 2, and so on round the wheel; the link needs 24 people.

Turn — a shift time, such as 06:20-13:45. An **early** starts before 09:00; a **late** starts after.

Cover (spare) week — a week with no fixed times. You work four duties of the seven days, placed by the roster clerk to cover leave and sickness.

The contract — 35 hours a week, averaged over the 24 lines; a cover week counts as a full contracted week.

Line — one week of the pattern, Sunday to Saturday. You move down a line each week, and line 24 goes back to line 1.

Rest day (RD) — a day off inside the contracted week. Two together make a proper break; one on its own does not count as a 48-hour break.

Sunday — not part of the contract. A Sunday duty is rest-day working, paid on top of the week, which is why the hours are shown Mon-Sat and all week.

One-turn week — the same clock time all week (shown bold in the Turns column); the easiest kind of week to live around.

The rules, and what fitting did

The ticket office is fixed; every other duty was chosen for how the floor follows the trains. How a week compares with today, and the duty table beside today's.

The shape of a week, against today

MEASURE	TODAY (20)	PROPOSED (24)
Working weeks that are one turn (one clock time Mon–Fri)	7 of 16	11 of 20
Weeks mixing early and late turns	7	4
Rest-day breaks that are two days or more	13 of 17	22 of 24
Single rest days between duties (not a 48h break)	4	2
Days worked in a week (lines × days)	1×3, 3×4, 11×5, 1×6	6×4, 14×5
Cover (spare) weeks (whole weeks)	lines 1, 7, 12, 17	lines 1, 7, 13, 19
Distinct shift times (all but 12 on today's roster)	18	17
Mon–Sat hours, average week (the contract)	35h 00m	35h 00m
Sunday duties (not contracted; half the working lines)	8 of 16 lines	10 of 20 lines

Changed — the December 2026 headcount

PEOPLE ON DUTY	TODAY	PROPOSED	WHY
Monday to Friday	11–12	14	the contract: 20 working lines × 35h has to be worked somewhere
Saturday	10	14	owner's figure for December 2026, evenly balanced to demand
Sunday	8	10	owner's figure for December 2026
Opening at 06:20 (07:15 Sunday) wk · Sat · Sun	4 · 4 · 3	5 · 5 · 4	at least four to open, every day — met
Through to the close (23:55; 23:25 Sunday) wk · Sat · Sun	2 · 2 · 3	4 · 4 · 3	at least three to close, every day — met
Still on duty at 22:00 wk · Sat · Sun	4 · 2 · 3	6 · 5 · 5	at least five from 22:00, every day — met

The duty table — the ticket office fixed, the floor fitted to the timetable

TURN	WK	SAT	SUN	WK	SAT	SUN
06:20–13:25 7h 15m · Early	4			4		
06:20–13:45 7h 25m · Early	1			1		
06:20–14:00 7h 40m · Early		4			4	
06:20–14:20 8h 00m · Early	2			2		
06:20–14:30 8h 10m · Early				1 +1		
06:20–14:50 8h 30m · Early		3			2 -1	
06:20–15:30 9h 10m · Early					3 +3	
06:20–15:45 9h 25m · Early				1 +1		
07:00–16:15 9h 15m · Early				2 +2		
07:15–15:30 8h 15m · Early						4 +4
07:15–15:45 9h 30m · Early						4
07:15–16:45 9h 30m · Early					2 +2	
08:00–16:30 9h 30m · Early	2			2		
08:30–16:30 8h 00m · Early			1			4
09:15–18:15 9h 00m · Early						1 +1
11:00–19:30 9h 30m · Late	4			4		
12:00–20:00 9h 00m · Late		4			4	
13:00–21:00 9h 00m · Late			4			4
13:30–21:00 7h 30m · Late		4			4	
13:30–22:00 8h 30m · Late	4			4		
13:30–22:30 9h 00m · Late						2 +2
14:00–21:00 7h 00m · Late				1 +1		
14:00–22:30 8h 30m · Late	2			2		
14:30–22:00 7h 30m · Late		2			2	
14:30–23:25 8h 55m · Late						3
14:45–22:45 8h 00m · Late				1 +1		
14:45–23:55 9h 10m · Late		2		2		
15:15–23:25 8h 10m · Late						3 +3
15:15–23:55 8h 40m · Late	2			2		
15:45–23:55 8h 10m · Late				4 +4	4 +4	

Busiest weekday · Saturday · Sunday. The first three columns are what the 20-line link does now; the last three are the proposal. Green cells grew, amber shrank, a struck-through row is a time the proposal does not use.

A weekday pays 6,970 minutes and a Saturday 7,150, so that five weekdays and a Saturday total exactly 20 × 35h = 42,000 — and the split itself was searched (page 9). Sunday pays 5,070, outside the contract. The ticket office is the fixed pairs (06:20–14:20 and 14:00–22:30 on a weekday, 06:20–14:50 and 14:30–22:00 on a Saturday, 07:15–15:30 and 13:30–22:30 on a Sunday); every other row was chosen for the floor's fit. 12 of the 17 times are ones nobody works today.

People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's traffic.

How to read it: each blue cell is the number of people on duty in that hour (darker blue = more people). The orange row is the timetable — how much of the day's trains that hour carries (darker orange = busier). Read the columns top to bottom: cover should be thickest where the orange is darkest. The **fit** figure at the right end of each row scores that match for the whole day.

Monday to Friday window 06:20–23:55 · one row per distinct weekday

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY MON · WED		4	4	6	6	6	7	7	7	7	7	7	5	5	5	4	4	4	2	55.8
TODAY TUE · THU · FRI		4	4	6	6	6	7	7	8	8	8	8	6	6	6	5	5	4	2	50
PROPOSED		5	7	7	7	7	7	7	7	9	10	9	7	7	7	7	6	6	4	21.9
OF WHOM TICKET OFFICE		2	2	2	2	2	2	2	2	4	2	2	2	2	2	2	2	2		
OF WHOM ON THE FLOOR		3	5	5	5	5	5	5	5	5	8	7	5	5	5	5	4	4	4	20.4
DEC 2026 TRAFFIC	5	69	107	127	122	95	85	75	84	79	87	105	140	138	107	90	73	75	66	

Saturday window 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY		4	4	4	4	4	4	5	6	9	6	6	6	6	6	5	4	2	2	45.7
PROPOSED		5	7	7	7	7	7	7	7	10	12	9	7	7	7	7	7	5	4	12.2
OF WHOM TICKET OFFICE		2	2	2	2	2	2	2	2	4	2	2	2	2	2	2	2			
OF WHOM ON THE FLOOR		3	5	5	5	5	5	5	5	6	10	7	5	5	5	5	5	5	4	11.6
DEC 2026 TRAFFIC	6	24	58	75	81	92	88	77	64	68	72	84	82	67	64	79	78	62	41	

Sunday window 07:15–23:25

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY			3	4	4	4	4	4	5	8	8	5	4	4	4	4	3	3	3	65.9
PROPOSED			4	4	5	5	5	5	7	7	10	6	6	6	5	5	5	5	3	33.6
OF WHOM TICKET OFFICE			2	2	2	2	2	2	4	4	4	2	2	2	2	2	2	2		
OF WHOM ON THE FLOOR			2	2	3	3	3	3	3	3	6	4	4	4	3	3	3	3	3	19.2
DEC 2026 TRAFFIC			15	45	65	71	74	71	61	55	64	69	78	73	75	52	71	65	50	

Spare cover is not in these figures — a cover week carries no times, so the rows are a floor. **Fit** is one number for how closely the people on duty follow the trains through the day: 0 would be a perfect match, lower is better, and today's link scores what it scores. (For the record: the squared gap between each hour's share of the day's traffic and its share of the cover, inside the day's window, measured on duty minutes — so an hour a duty only touches counts in proportion; the cells above count heads.) It is the measure the searched tables were chosen on, and the fit of the average weekday is the *Wk fit* on page 9. Sunday: 5 December 2026 timetable movements fall after the 23:25 finish (the last at 23:54, three of them arrivals) — the standing question on whether Sunday's window moves; the window is stored per design, so the proposal can be rebuilt to either answer.

Reading it. The two indented rows split the proposal: the ticket office is the fixed pairs, the floor is everyone else, and the floor's fit is the figure the day tables were chosen on (20.4 · 11.6 · 19.2). The fit measures **shape, not level** — each hour's share of the day's cover against its share of the day's trains — so more people in a quiet hour makes it worse even though nobody is worse off. The ticket office is staffed to its opening hours, not to the trains, which is why the overall fit on the proposed row is looser than the floor's. Saturday was fitted evenly to the timetable, with no weighting for events.

The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

Hard limits — a design either meets these or cannot be run

- ✓ **More than 13 consecutive days worked** — longest possible run **6** days (today: 9)
Longest possible run is 6 days, within the 13 configured here from Chiltern practice (origin: the legacy Hidden standard). A spare week is 4 duties of 7 and the roster clerk places them, so this is the worst case the link allows. Worked as a BLOCK of four rather than split day-on-day-off, the answer is the same — **6** days — so this row does not depend on how a cover week is placed.
Basis: Chiltern roster policy, citation outstanding — legacy Hidden 13-in-14 standard. Configured from Chiltern practice; the policy citation is outstanding, so this is stated as the app states it.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)
Tightest anywhere: **14h 05m — line 2 Fri 07:00-16:15 into Sat 06:20-14:50** (today's tightest: 12h 30m). The generator refuses a design it cannot repair to this; the search here never produced one.
- ✓ **The contracted week, exactly** — **35h 00m** average Mon-Sat over the 24 lines, cover weeks counted as contracted weeks
700h of duty a week across 20 working lines. Sundays (84.50h) sit on top as RDW, as they do today. Individual weeks range 24h 30m–46h 55m; only the average is the contract.

December 2026 timetable design figures — the staffing shape agreed for the new timetable

RULE	PROPOSAL	
At least four on at the open, every day	✓ 5 weekday · 5 Saturday · 4 Sunday	a minimum
At least three through to the close, every day	✓ 4 weekday · 4 Saturday · 3 Sunday	a minimum
At least five on duty at 22:00, every day	✓ 6 weekday · 5 Saturday · 5 Sunday	a 22:00 finish is not "on at 22:00"
Fourteen on a Saturday, ten on a Sunday	✓ 14 · 10	
Four cover weeks, evenly spread	✓ lines 1, 7, 13, 19 — gaps 6, 6, 6, 6	whole weeks, never scattered days
The ticket office, fixed pairs of identical turns	✓ ✓ Mon-Fri 2× 06:20-14:20 · ✓ 2× 14:00-22:30 · ✓ Sat 2× 06:20-14:50 · ✓ 2× 14:30-22:00 · ✓ Sun 2× 07:15-15:30 · ✓ 2× 13:30-22:30	not counted as floor cover
Every weekday closer starts 15:45	✓ 4× 15:45-23:55	was 15:15
At least two on the floor at every moment	✓ fewest 3 weekday · 3 Saturday · 2 Sunday	every five minutes, the office left out
Handovers — 15 minutes to each closer, 20 in the ticket office	✓ floor ✓ ✓ ✓ · office overlap 20 · 20 · 120 min	Sunday openers stay 15 min past the last closer's start
Sunday duties 8h to 9h; every other day 7h to 9h30	✓ Sunday 8h10-9h00 · Mon-Sat 7h00-9h30	
No more shift times than today	✓ 17 against today's 18	
About 4.2 days a week worked, Mon-Sat	✓ 4.20 over the 20 working lines	Sunday is outside the 35-hour average

To settle before it is frozen

Weekday against Saturday. Five weekdays and a Saturday pay exactly 42,000 minutes, so each split trades one against the other; every split was proven (floor fit weekday / Saturday): 6,960 → 20.2 / 12.9 · **6,970 → 20.4 / 11.6, this sheet** · 6,975 → 20.6 / 10.8 · 6,990 → 20.9 / 9.2. **The fit is share-based** — it scores shape, not level, so read it beside the heads on page 6. **Floor help from the office** — the second office person on the floor before 08:00 and after 19:30 — is not modelled here, and would move the best table. **Sunday's finish** — 5 December 2026 timetable movements fall after 23:25; inherited, not decided.

ORR fatigue factors

Good practice guidelines — Fatigue Factors, p3 (December 2021). Δ present — the pattern is in this design and worth a look, not a breach · $\checkmark$ clear — it is not · $\bullet$ standing — true of the station itself, not of any design · — does not apply here. This is the list the link is assessed against.

The Office of Rail and Road lists 25 roster patterns that tend to tire people — long runs of earlys, short gaps between duties, start times that jump about. For each one this table asks whether the pattern is in the rotation, today and proposed, and how big it is. The list is guidance: a factor present is a question for the room, never a pass or a fail, and a design showing nothing is not thereby approved.

CODE	FACTOR	TODAY'S LINK	PROPOSED
FF1	Night shift covering 00:00–05:00 Time of day	--	--
FF2	Early shift starting 05:00–07:00 Time of day	$\bullet$ 24	$\bullet$ 30
FF3	Very early shift starting before 05:00 Time of day	$\checkmark$ 0	$\checkmark$ 0
FF4	Very early shift (before 05:00) over 8h Duty length	--	--
FF5	Day shift over 12h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF7	Early shift over 10h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF8b	Fewer than 2 days rest after a block of early starts Recovery time	Δ 3	$\checkmark$ 0
FF11	More than 13 consecutive shifts without a 48h break Recovery time	$\checkmark$ 12 $\checkmark$ 9 as rostered	$\checkmark$ 9 $\checkmark$ 9 as rostered
FF10	More than 4 consecutive 12h day shifts Cumulative	$\checkmark$ 0	$\checkmark$ 0
FF15	More than 4 consecutive early shifts in a rotating pattern Cumulative	Δ 6	$\checkmark$ 4
MRSF	More than 12 consecutive day shifts Cumulative	$\checkmark$ 9	$\checkmark$ 6
MRSF	More than 55 hours worked in any 7-day period Cumulative	Δ 60.8	$\checkmark$ 53.5
MRSF	More than 7 consecutive 8h shifts (definition to confirm) Cumulative	Δ 9	$\checkmark$ 6
MRSF	More than 6 consecutive night or early shifts in a permanent pattern Cumulative	--	--
FF17	Backward rotating pattern (definition to confirm) Circadian	$\checkmark$ 10 backward / 12 forward	$\checkmark$ 14 backward / 14 forward
FF18	Rotating pattern of about a week (definition to confirm) Circadian	$\bullet$ 20-line rotation · typically 4h 0m a week	$\bullet$ 24-line rotation · typically 1h 50m a week
FF19	Successive start times varying by more than 2 hours (definition to confirm) Circadian	Δ 6	$\checkmark$ 0
FF6	Night shift over 10h Night shifts	--	--
FF8	Less than 2 days rest after consecutive nights Night shifts	--	--
FF9	Less than 14h rest in any 24h for night shifts Night shifts	--	--
FF12	Only one day rest after night shifts Night shifts	--	--
FF14	More than 4 consecutive nights in a rotating pattern Night shifts	--	--
FF16	More than 3 consecutive night shifts over 8h Night shifts	--	--
FF20	First night shift Night shifts	--	--
FF13	Less than 12h rest in any 24h Intervals between duties	$\checkmark$ 0	$\checkmark$ 0

Present: 0 (today 5) · standing: 2 · FF2 fires on every 06:20 duty, so it is a property of the station's opening time rather than of any design. **Two readings of a cover week, on the one row they move.** A cover week is worked four days of seven and the link does not say which four, so a run figure is a range. The headline number is the ceiling — the four split day-on-day-off, which supplies no 48-hour break and joins the blocks either side. It is reachable: of the 35 ways to place four duties in seven days, the 10 that leave no two rest days together produce exactly it. Worked as a BLOCK, which is what a cover week looks like on the roster, the three rest days fall together and the week always supplies a break — that is the *as rostered* figure beneath it. Checked on every row: FF11 is the only one where the two differ. Which reading applies is a question for the roster office, not for this sheet. This sheet is an aid to a conversation, not a fatigue risk assessment.

How it was chosen

Reproducible: the same table, seed and objective give the same grid on any machine

Lineage. Family FR — built to the owner's final rules of 28 September 2026. The ticket office is written in first as fixed pairs of identical turns and is not counted as floor cover: Monday to Friday two 06:20–14:20 and two 14:00–22:30, Saturday two 06:20–14:50 and two 14:30–22:00, Sunday two 07:15 earlies and two lates to 22:30. Every weekday closer starts 15:45. The rest of each day was enumerated to a proof for the closest fit of the floor to the December 2026 timetable, with the headcounts as minimums, at least two on the floor at every moment, handovers of 15 minutes to each closer and 20 in the ticket office, Sunday duties of 8h to 9h, and no more shift times than today. The rotation was then searched for the fewest fatigue factors. Compare with “Pinned Turns 2” (P2-24-N13 · 33a78cbe), “Pinned Turns” (PT-24-P34 · dae6292e) and “By the Book” (BB-24-D7 · 0f14abce).

In plain terms. A computer took each day's duties as fixed and tried over a million different ways of arranging them across the 24 weeks, scoring every arrangement on the rules on pages 7 and 8, and kept the best. The “seed” is the number that makes that run repeat exactly on any machine, so anyone can rebuild this grid and get the same cells.

How it was chosen

1 • The table. final-table.mjs fixed the ticket-office pairs and the 15:45 closers first, then enumerated every remaining table on the five-minute grid — each duty 7h–9h30 (8h–9h on a Sunday), starting at the window's own instant or on the quarter hour — keeping only tables that meet every minimum, keep two on the floor at every moment, hand over 15 minutes to each closer and 20 in the ticket office, and use no more shift times than today. Branch and bound kept the table whose floor, with the office's minutes taken out, follows the December 2026 timetable most closely. The weekday and Saturday minutes were swept in pairs that pay exactly 42,000 a week and each pair proven; 6,970 and 7,150 were chosen (page 7 gives the frontier).

2 • The grid. Simulated annealing over the 24-line grid: 100,000 candidate arrangements per restart, five restarts from different random starting points per run, four seeded runs, fatigue-first — over a million candidates in all — then a steepest-descent polish of every

same-day swap and every pair of lines. Every move keeps each day's duties exactly as the table says — a swap between two lines on one day, or two whole lines changing places — so the coverage, the contract and the headcounts are true by construction and only the SHAPE is searched. Cover weeks stay at lines 1, 7, 13 and 19 throughout.

3 • The judge. Every candidate was scored by the workspace's own modules — runDesignChecks, assessFatigue, assessHardLimits, scoreOrder — with a rest under 12h or a run past six days costing more than everything else combined, fatigue factors next, and then the feel terms: a week on more than one turn, a single rest day between duties, fewer than four or more than five days, and the week-to-week step. Weekends off and long weekends were rewarded. Last, the app's line-order optimiser was offered the result and kept only where it improved that score.

Alternatives measured

DESIGN	RUN	FF PRESENT	WKENDS	ONE-TURN	STEP	WK FIT	SCORE
FR-24-F34 · 03a59c77 — Final Rules (this proposal)	6	0	6	11/20	1h 50m	21.9	12700
FR-24-F21 · e5969aab — runner-up	6	0	5	12/20	2h 51m	21.9	12410
Workspace default · 29ec9810 (Dec 2026 table, generated)	7	5	5	0/20	4h 16m	30.8	2192436
P2-24-N13 · 33a78cbe — Pinned Turns 2 (a sibling proposal)	6	0	6	16/20	2h 18m	33.8	—
PT-24-P34 · dae6292e — Pinned Turns (a sibling proposal)	6	0	6	14/20	2h 9m	32.1	—
BB-24-D7 · 0f14abce — By the Book (a sibling proposal)	6	0	6	12/20	2h 52m	30.8	—
Today's 20-line link (for scale)	9	5	4	7/16	4h 0m	51.1	—

How the winner was picked: rules first (rest, run, fatigue factors present, weekends off), then *Wk fit* — how evenly the weekday cover follows the December 2026 timetable traffic curve (lower is better; today's link scores what it scores). Full weekends off decided it (6 against 5), and the runner-up is named so it can be asked for. The other 2 seeded results are in results/ (F13, F7); none stands higher on the pick than the rows shown. Score is the search's own feel objective (lower is better) and is not a verdict. The workspace default is the app's own December 2026 duty table, generated and reordered with every switch on — a good design by every panel, and not one that resembles today's.

LOAD IT

The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

Links → Import, then paste the whole block below. Line number, then Sunday to Saturday; SP is a cover week. The workspace re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust — and the two designers can edit it there like any other design.

The same rotation is also supplied beside this PDF as Final-Rules-FR-24-F34-import.txt (tab-separated, pastes directly) and Final-Rules-FR-24-F34.json (the app's own format).

1	SP	SP	SP	SP	SP	SP	SP
2	13:30-22:30	14:00-22:30	RD	RD	07:00-16:15	07:00-16:15	06:20-14:50
3	07:15-15:30	06:20-14:20	RD	RD	06:20-14:20	07:00-16:15	06:20-15:30
4	RD	RD	06:20-14:30	07:00-16:15	06:20-14:30	06:20-14:30	06:20-15:30
5	RD	RD	06:20-14:20	06:20-14:20	06:20-14:20	06:20-14:20	RD
6	RD	06:20-13:45	06:20-13:45	06:20-13:45	RD	RD	07:15-16:45
7	SP	SP	SP	SP	SP	SP	SP
8	RD	RD	15:45-23:55	15:45-23:55	15:45-23:55	15:45-23:55	RD
9	RD	07:00-16:15	06:20-15:45	06:20-15:45	06:20-15:45	06:20-15:45	RD
10	RD	06:20-14:30	07:00-16:15	06:20-14:30	07:00-16:15	RD	06:20-15:30
11	07:15-15:30	06:20-14:20	06:20-14:20	RD	RD	06:20-14:20	07:15-16:45
12	07:15-15:30	07:00-16:15	07:00-16:15	07:00-16:15	RD	RD	14:30-22:00
13	SP	SP	SP	SP	SP	SP	SP
14	09:15-18:15	RD	RD	06:20-14:20	06:20-13:45	06:20-13:45	06:20-14:50
15	RD	RD	14:00-22:30	14:00-22:30	14:00-22:30	14:00-22:30	14:45-22:45
16	13:30-22:30	RD	RD	14:00-22:30	14:00-22:30	14:00-22:30	RD
17	RD	14:00-21:00	14:00-21:00	14:00-21:00	RD	RD	15:45-23:55
18	15:15-23:25	15:45-23:55	15:45-23:55	RD	RD	15:45-23:55	14:30-22:00
19	SP	SP	SP	SP	SP	SP	SP
20	07:15-15:30	06:20-15:45	RD	RD	14:00-21:00	14:00-21:00	15:45-23:55
21	15:15-23:25	15:45-23:55	RD	RD	15:45-23:55	15:45-23:55	RD
22	RD	15:45-23:55	15:45-23:55	15:45-23:55	15:45-23:55	15:45-23:55	RD
23	RD	14:00-22:30	14:00-22:30	15:45-23:55	15:45-23:55	RD	15:45-23:55
24	15:15-23:25	15:45-23:55	15:45-23:55	15:45-23:55	RD	RD	15:45-23:55

Prepared 28 September 2026. The code **03a59c77** in every footer is the design's fingerprint — eight characters computed from these 168 cells and nothing else. If the workspace shows the same code after you paste, you have exactly this design; if one cell differs, the code changes. Every figure in this document was computed by the Marylebone Roster app's Links modules (v24.35) from exactly these cells; import them and the workspace's Design checks, hard limits, fatigue factors and coverage cards will restate them.